

Structure Extraction from PDF-based Book Documents

Liangcai Gao, Zhi Tang
Institute of Computer Science
& Technology, Peking
University, Beijing, China
{gaoliangcai,
tangzhi}@icst.pku.edu.cn

Xiaofan Lin
Vobile Inc.
4699 Old Ironsides Drive,
Santa Clara, California, USA
xiaofan@vobileinc.com

Ying Liu
Department of Knowledge
Service Engineering, KAIST,
Daejeon, Republic of Korea
yingliu@kaist.ac.kr

Ruiheng Qiu
State Key Laboratory of Digital
Publishing Technology, Peking
University Founder Group
Co., Ltd., Beijing, China
chourh@founder.com.cn

Yongtao Wang
Institute of Computer Science
& Technology, Peking
University, Beijing, China
wangyongtao@icst.pku.edu.cn

ABSTRACT

Nowadays PDF documents have become a dominating knowledge repository for both the academia and industry largely because they are very convenient to print and exchange. However, the methods of automated structure information extraction are yet to be fully explored and the lack of effective methods hinders the information reuse of the PDF documents. To enhance the usability for PDF-formatted electronic books, we propose a novel computational framework to analyze the underlying physical structure and logical structure. The analysis is conducted at both page level and document level, including global typographies, reading order, logical elements, chapter/section hierarchy and metadata. Moreover, two characteristics of PDF-based books, i.e., style consistency in the whole book document and natural rendering order of PDF files, are fully exploited in this paper to improve the conventional image-based structure extraction methods. This paper employs the bipartite graph as a common structure for modeling various tasks, including reading order recovery, figure and caption association, and metadata extraction. Based on the graph representation, the optimal matching (OM) method is utilized to find the global optima in those tasks. Extensive benchmarking using real-world data validates the high efficiency and discrimination ability of the proposed method.

Categories and Subject Descriptors

H.3.3 [Information Systems]: Information Search and Retrieval

General Terms

Algorithms

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JCDL '11, June 13–17, 2011, Ottawa, Ontario, Canada.

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Keywords

Electronic Book, Structure Extraction, Layout Analysis, Bipartite Graph

1. INTRODUCTION

In the digital era, an increasing number of documents are available in PDF format due to its excellent printing and exchange capability. However, PDF-based documents themselves lack information about document structures, which include two types: physical structure (line, block, column, spatial relationship, etc.) and logical structure (heading, figure caption, table, chapter/section hierarchy, reading order, etc.). Despite the fact that the latest PDF specifications make it possible to embed the structural information, most PDF producers do not make use of such features but still focus on the visual representation of the documents. Consequently, lots of valuable information related to physical and logical structures is lost, which limits the possibility of automatic information retrieval and reuse in many ways. For example, it is unlikely to process structural queries targeting academic PDF documents, such as “Publisher = ‘ACM Press’” or “Title LIKE ‘%Structured documents%’”, since the logical information is not present in those documents.

There are a number of established products on the market for semi-automatic navigation and extraction of data from Web pages. These solutions make use of the inherent structure of HTML to locate instances of data to be extracted. As PDF documents do not have such structures, extracting information from PDF documents has long been known to be challenging. Thus, a key motivation of our research¹ is to convert the display-oriented PDF documents to a structured format in order to support the wide range of electronic book applications, such as structure-based retrieval, reformatting, and printing-on-demand.

Document structure information extraction, as a topic in document analysis and understanding, has attracted much attention from researchers since the 1980s [30]. Most of existing research on this task has been focusing on the

¹Sponsored by China Postdoctoral Science Foundation (No.20100480125)

image-based documents. Structure extraction from PDF documents is still an emerging topic and state-of-the-art methods usually adopt the image-based features instead of the intrinsic characteristics of PDF format. In this paper, we will make full use of those characteristics to improve the conventional image-based method. The logical structure of a document varies with the document type. As Lee et al. [20] state, the development of a general structure analysis method for all document types is nearly impossible. Therefore, existing approaches are normally domain-specific and effective only for the targeted type of documents, such as software specification, mail pieces, scientific journals, and newspapers. Regarding the structure information extraction from book documents, existing research focuses on its specific subtasks, and none of them provides a generic solution. Furthermore, those methods are usually sensitive to the layout variability. Namely, a method effective for one type of books often fails on the other types of books with different layout formats.

Compared with the previous works, this paper aims at book documents in PDF format and proposes a Structure Extraction from Books (SEB) system to extract their major structure information. After analyzing the characteristics of book documents and PDF files, this paper presents a set of new methods to improve the performance of some specific structure extraction tasks based on bipartite graphs, clustering, and a common typesetting practice in books: within a book, page objects belonging to the same functional component usually share format regularity.

This paper is organized as follows. Section 2 summarizes related research on structure extraction from PDF book documents. Section 3 analyzes the characteristics of book documents and PDF files, and illustrates our basic idea of book structure information extraction. Sections 4, 5, and 6 present the details of the proposed framework, including identifying physical structure shared by multiple pages of a book (known as global typography detection), logical structure extraction at the page level (logical element labeling and reading order detection), logical structure extraction at the document level (chapter/section hierarchical structure extraction and metadata extraction). Section 7 discusses the experimental results and Section 8 draws conclusions.

2. RELATED WORK

As a very popular application of structure extraction from PDF documents, pdf-to-html conversion has been worked on for years and some tools have been developed recently [15, 28]. A survey on extracting low-level structural data from PDF documents can be found in [16]. Besides, several relevant techniques of PDF document analysis and understanding have been proposed in recent years. For example, Anjewierden [3] describes a document analysis system, AIDAS, which incrementally extracts logical structures from PDF documents. However, the grammars used in the system depend on the specific document types (e.g., technical manuals). Rigamonti et al. [28] demonstrate a reverse engineering tool for PDF documents, named Xed, which is able to extract the original document’s physical structure in a hierarchical canonical form independent of the document type. Bloechle et al. [5] propose Dolores, an interactive and class-free approach for PDF document logical restructuring, which includes the

learning capability to recover the logical structures from various documents such as newspapers, books, scientific papers, and journals. Déjean et al. [11] present a system for converting legacy PDF documents into structured XML format. This conversion system first extracts the heterogeneous streams contained in the PDF files and then transforms the logically structured documents into XML format through a series of steps. Most of these operations are based on the conventional document image analysis, while some are specific to PDF documents. Hassan [16] develops a system for wrapping PDF documents, in which a PDF document is represented as an attributed relational graph. A wrapper is defined as a sub-graph of the document with additional constraints. After that, an algorithm based on sub-graph isomorphism is used to find the data instances and extract the related data. Nojournian et al. [26] aim at document engineering of PDF-based software specifications. The first step is to extract the logical structure of the document into a meaningful XML format. Then it creates the multi-layer hypertext version of the document.

Some work focuses on the structure extraction from book documents. Lin et al. [21] propose a method for analyzing the logical structure of books based on their tables of contents (called ToC hereafter) by layout modeling and headline matching. The structures of page number, header, footer, headline, figure and body text are analyzed and matched with information on the contents pages to reconstruct the links between ToC and body text. He et al. [17] propose a method to extract the hierarchical logical structure of book documents, along with the reference information, by combining the spatial and the semantic information of ToC in a book.

3. BASIC IDEA

We utilize the following three characteristics of PDF-based books to finish structure extraction, combining the image-based features used in precious works.

First, as Bart et al. [4] point out, repetition of physical structure is prevalent in documents. Such repetition conveys the underlying logical structure of the data in document design. Within a book, the repetition is manifested as a common typesetting practice that page elements belonging to the same functional component usually share format regularity, which we call “Style Consistency of page Components in the whole book”, or SCC. Here the notation “component” refers to the specific logical structure in a page, such as heading, table, and figure caption. In the following sections, an “element” or “block” refers to the geometric segment of a page, such as part of text, image, and graphics. SCC can be considered as a special relation between the different elements belonging to the same component throughout the whole book. For example, the headings, footnotes and citations at the same level in a book usually adopt the same format.

As a common typesetting practice, SCC has been long advocated by many leading figures in publishing, such as Jan Tschichold² and Stanley Morison³. They have designed some classic page layout styles for books. For example, the Penguin Composition Rules designed by Jan Tschichold in 1947 had been adopted by millions of Penguin books

²http://en.wikipedia.org/wiki/Jan_Tschichold

³http://en.wikipedia.org/wiki/Stanley_Morison

[9]. Some researchers have taken advantage of SCC in document structure extraction tasks. For example, Déjean et al. [11] utilize the fact that the entries at the same level in ToC share the same visual characteristics to determine the hierarchy of ToC entries. Lin [22] detects headers and footers according to the consistency of headers and footers across different pages in multi-page documents. In order to extract such repeated structures, Bart and Sarkar [4] propose an example-based matching method to locate additional instances of the example structure in multiple documents. As surveyed by Shipman et al. [29], people use the visual and spatial characteristics of graphical layouts to express relationships between objects. And they discover a small set of underlying primitives that capture the conventions and implement a set of heuristic algorithms to recognize the common implicit spatial structures. In this paper, we adopt SCC as an important clue of structure extraction from books. The details are introduced in the following sections.

Second, as a display-oriented format, a PDF document stores a set of rendering instructions, which is the meta-information about the physical structure of a document used to specify where and how each text, graph or image unit should be printed on a page. Thus, the fonts, characters, and coordinates can be accurately extracted from PDF documents by many off-the-shelf PDF converters. Compared with image files, PDF files could provide richer and more reliable information, especially textual content, which can be used to improve the conventional image-based structure extraction methods [1]. In our method, the content information is deeply mined to accurately establish the relationship between page components.

On the other hand, in modern publishing, word processors, including PDF generators, often render the content as a postscript stream in the logical reading order. Therefore the natural rendering order of content in a PDF document usually reflects the logical reading order. This natural order has been used in some pdf-to-html tools [15]. However, since PDF files are generated to accurately represent the document’s visual appearance, keeping the internal structure representation is not mandatory. Occasionally surprising results might appear in the output stream of a PDF file [28]. For instance, a single word missing in a paragraph might show up as an isolated word at the end of the document. More commonly, the block order across the pages might be confusing because the blocks may not be always rendered in the logical reading order. The above observations show that the natural rendering order of the content in a PDF document is not always reliable and other information (e.g. spatial relationship) is needed to determine and validate the logical reading order. In this paper, we investigate how to better employ this feature in book structure extraction through leveraging the blocks’ visual appearance.

Finally, this paper models several structure extraction tasks as a one-to-one matching problem, describes them using the bipartite graph, and employs the optimal matching (OM) based on Kuhn-Munkres algorithm [27] to get the globally optimal matching, which outperforms the existing work. The following sections discuss the details.

4. SYSTEM ARCHITECTURE

In terms of functionality, this paper introduces a reverse engineering system of document typesetting: the structure

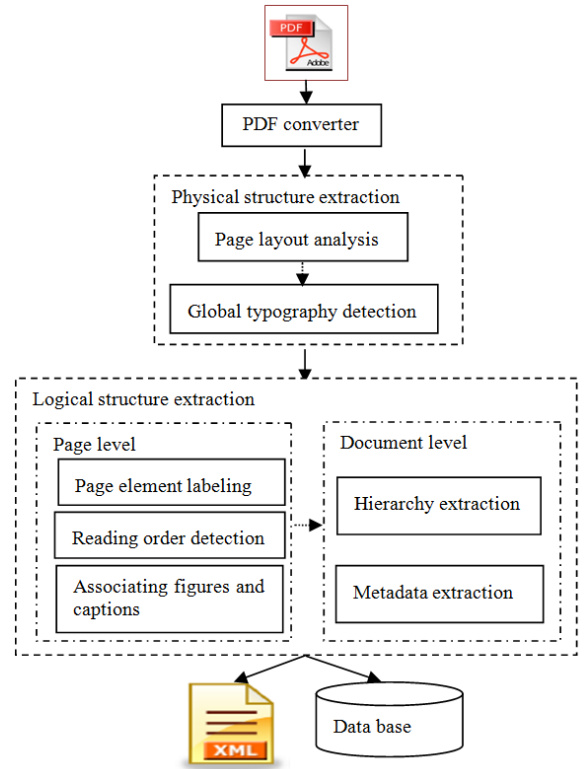


Figure 1: SEB System architecture

extraction from PDF-based books system, SEB. This system includes two parts: physical structure extraction and logical structure extraction, as shown in Figure 1.

Different pages within a book usually share layout typographies, such as the fonts of body text/header, header/footer, column, text direction, and line spacing. We define such stable typographies throughout a document as global typographies. There are two levels of logical structures within a book, i.e., page level and document level. The former includes the reading order, and logical page component labels such as heading, figure/table caption, and footnote. And the latter includes the document chapter/section hierarchy structure and metadata in the citation data and title page. Our system architecture covers the major tasks of structure extraction in two levels.

5. PHYSICAL STRUCTURE EXTRACTION

5.1 Page Layout Analysis

We do page layout analysis by merging small page objects (e.g. characters, words, lines) into big ones in a bottom-up manner. The details are introduced in Step b) and Step c) of Section 6.1.2.1 since they are implemented inside the reading order detection component.

5.2 Global Typography Detection

Global typographies extracted in the system include columns, header and footer, page body area, text line directions, line spacing of body text, and fonts used in various components (headings, paragraphs, etc.). Such information is consistent throughout a multi-page document, especially in a book document. Thus they can

be considered as a part of physical structure in document level and an embodiment of SCC in page layout structure. Extracting such global typographies will benefit document analysis at both the document and page levels. For example, header and footer recognition helps to spot page numbers by narrowing down the search range. Furthermore, if we know the fonts used at different levels of headings, the titles of chapters or sections can be more easily extracted, and the ToC could be regenerated as well. In addition, the required storage of the global typography information is low, since it is expressed in the document level rather than for each page.

One significant advantage of the proposed method is the utilization of page association. For example, to detect headers and footers, a similarity score is calculated through matching the text and geometric position between the top/bottom lines on neighboring pages. Columns are identified by detecting the recurring white spaces in multiple pages. Furthermore, the PDF file’s natural rendering order is also used in this task. For example, text direction is detected by comparing the coordinates of the neighboring characters in the natural rendering order rather than segmenting a page into lines. More details in this regard can be found in our previous paper [13].

6. LOGICAL STRUCTURE EXTRACTION

This section discusses logical structure extraction at both page level (Section 6.1) and document level (Section 6.2).

6.1 Page Level Extraction

6.1.1 Page Element Labeling

Page element labeling aims to assign a logical label (e.g., text paragraph, headings, footnote, figure/table caption) to each block of a page. It can be considered as a classification problem [2]. A number of rule-based or learning-based methods have been proposed for the problem [32]. Normally the features used in those methods are based on the characteristics of the text blocks within a page. Although we cannot expect consistent typesetting for a certain type of components in different books, it is common that text blocks related to the same component type have the same visual characteristics within the same book. This is one example of SCC as defined in Section 3. Consequently the relationship of the blocks belonging to the same component type in different pages can be used to group blocks. For example, if a font is always adopted in the text blocks under the images in a book, it is very possible that the text blocks adopting the same font are figure captions. SEB first groups the text blocks of a document based on SCC via clustering. The output is a set of clusters which correspond to different logical component types and serve as the prototypes of similar blocks. After that, the system uses a learning-based classification method to label the blocks in each cluster. The statistical and formal features used in SEB are as follows:

Headings: In terms of appearance, the text blocks from the same heading cluster often occupy one or two lines and adopt large font sizes. Especially, the text blocks in the chapter heading cluster usually appear in specific positions of a number of pages throughout the whole book, because books usually begin a new chapter on a new page and arrange chapter headings in a fixed position of the page. In addition, since the number of headings is much smaller

Table 1: Features of a block

Features	Meanings
Features in block level	
<i>FontStyle</i>	Font style used in a block, e.g. bold, italic
<i>FontSizePlace</i>	The place of font size in the whole fonts used in a document
<i>BlckLineNum</i>	The number of lines
<i>BlckWorkNum</i>	The number of words
<i>KWDFigure</i>	Begin with the keywords of figure caption
<i>KWDTable</i>	Begin with the keywords of table caption
<i>KWDFootnote</i>	Begin with the keywords of footnote
<i>NumSeqBegin</i>	Begin with the numeral expression
<i>SpncEnd</i>	End in punctuation
<i>PagBottom</i>	Located at the page bottom
<i>PagTop</i>	Located at the page top
Features in page level	
<i>BlckToBlkDist</i>	Ratio of the distance between blocks to line spacing
<i>Imgneighbor</i>	Adjacency with images
Features in document level	
<i>ClstSize</i>	The number of blocks contained in a cluster
<i>ClstToFigNum</i>	Ratio of the cluster size to figure number
<i>CBlckLinNum</i>	Consistency of blocks in <i>BlckLineNum</i>
<i>CKWDFigure</i>	Consistency of blocks in <i>KWDFigure</i>
<i>CKWDTable</i>	Consistency of blocks in <i>KWDTable</i>
<i>CKWDFootnote</i>	Consistency of blocks in <i>KWDFootnote</i>
<i>CNumSqnceB</i>	Consistency of blocks in <i>NumSeqBegin</i>
<i>CSpncuationEnd</i>	Consistency of blocks in <i>SpncEnd</i>
<i>CPagBottom</i>	Consistency of blocks in <i>PagBottom</i>
<i>CPageTop</i>	Consistency of blocks in <i>PagTop</i>

than the number of ordinary text paragraphs, the size of the heading cluster is usually small.

Figure/table captions: The size of figure/table caption cluster should be roughly equal to the number of figures/tables in a document. Geometrically, the text blocks belonging to the figure/table caption cluster should surround the figures/tables. In addition, they often contain several common keywords such as “figure”, “table”, “picture”, and “graph” [23].

Footnotes: The text blocks belonging to footnote cluster appear at the bottom of pages. Furthermore, they appear in a relatively small font, and they often contain numbers or certain unique tags (e.g., “*”) at the beginning of the text blocks.

Based on the above observations, SEB extracts 23 features from each block. As shown in Table 1, they are at three levels: block level, page level and document level. The block level features reflect the local attributes of a block. The page level features reflect the relationship between a block and its neighboring blocks within a page. The document level features reflect the local attributes of the cluster containing the block. And a learning method, Support Vector Machine (SVM), is employed to label the text blocks based on all of those features.

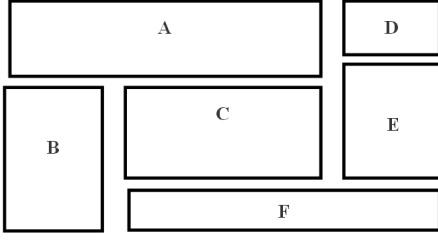


Figure 2: An example of L-shapes

6.1.2 Reading Order Detection

The conversion from the display-oriented PDF format to a structured format is of particular interest to us. A critical task in the conversion is to determine the logical reading order of the elements on each page. For example, the reading order of a page shown in Figure 2 is A-B-C-D-E-F.

Some ordering approaches have been proposed. They exploited geometric features of the page objects, and even the content of objects, with or without a *priori* knowledge for a particular document class. And the most common technology is the XY-cut algorithm, first proposed by Nagy [25] as a page segmentation method. And an XY-cut means a horizontal/vertical partitioning of the input image. Ishitani et al. [18] and Meunier [24] respectively propose an optimized XY-cut algorithm for determining a page reading order through an elimination-insertion procedure or a global cutting strategy. But the L-shapes (shown in Figure 2) still pose a challenge for the XY-cut algorithm. Correspondingly, some methods beyond XY-cut are also proposed. For example, Breuel [6] and Aiello et al. [1] recover reading order using topological sorting and spatial reasoning. And a lexical analysis technique is also adopted in [1] to score reading orders. Some learning-based methods such as inductive logic programming, probabilistic Bayesian classifier and preference learning are also used to identify a partial ordering of page elements [8].

6.1.2.1 Algorithms.

In this paper, we adopt the following steps to improve existing methods according to the characteristics of PDF-based book documents and a graph-based algorithm.

Step a): Page preprocessing. In order to keep the continuity of the body text, we first eliminate headers and footers detected in Section 5. In the output stream of a PDF file, the text and image object belonging to the same object might be independently output, and the separated text may disturb the ordering of other text. So the text and image objects with overlapping bounding boxes are combined into a compound object before ordering.

Step b): Find the separation lines. The graphical lines in a page are often used to visually separate different parts on the page. And our approach enforces cuts along lines and discards cuts crossing those lines. This step is straightforward since PDF converters output those lines in SVG format.

Step c): Merge small blocks into big ones. Most of PDF converters can provide bounding boxes of page objects. Our approach merges small objects (e.g. characters, words, lines) into big ones in a bottom-up manner, given the following conditions:

- 1: The positional relationship between the two objects is adjacent, the distance between them is less than a

threshold, and their geometric relationship conforms to a predefined model.

- 2: The two objects have the same text line orientation.
- 3: As another example of SCC, the most frequent font used in two text objects is the same. In other words, two objects adopting different fonts usually belong to different components and should not be merged.
- 4: The two objects are adjacent in the output stream of PDF files. This condition is based on the local reliability of PDF’s natural rendering order. Although it is possible that natural order is not reliable within a page in some PDF files, within a block the natural order is reliable in most PDF files.

Step d): Order those blocks generated from Step c) by XY-cut algorithm. Our approach first cuts a page on the column position calculated in Section 5, because different pages of a book usually share the same column layout. When L-shapes appear in a page, our approach first selects the blocks to be eliminated, which span more than one column. This is based on one observation: the blocks causing L-shapes in a page often span more than one column and usually are not the body text blocks. The reading order generated from the XY-cut is simply referred as “Cut-Order” henceforth.

Step e): Order the blocks by optimal matching (OM). The reading order determination is to induce an order between blocks of a page, and can be seen as finding the only successive block in reading order for each block of a page. So this problem can be considered as a one-to-one matching problem. In the proposed approach, a weighted bipartite graph [31] is constructed to model those blocks and their probability as neighbors in the reading order. Then the Kuhn-Munkres algorithm is employed to compute the optimal matching of the constructed bipartite graph. And the reading order is generated from the matching. It is obvious that this OM-based ordering method does not depend on XY-cut, thus it can avoid the problem of L-shapes. In addition, matches through OM reflect a global optimum of block ordering.

6.1.2.2 Constructing Weighted Bipartite Graph.

In a bipartite graph, all the vertices of the graph are split up into two complementary parts. And each edge of the graph connects two vertices respectively in the two parts. Namely no edge connects two vertices belonging to the same part. A “matching” of the graph is a sub-graph, in which each vertex only is connected by one edge. The matching with the maximal edge weights is defined as the optimal matching.

The weighted bipartite graph of those blocks in a page is constructed as follows:

Let $\mathbf{X} = \{x_1, x_2, \dots, x_p\}$ as a page with p blocks, and x_i represents a block in $\mathbf{X}$.

Let $\mathbf{Y} = \{y_1, y_2, \dots, y_q\}$ as a duplicate of $\mathbf{X}$.

Let $\mathbf{G} = \{\mathbf{X}, \mathbf{Y}, \mathbf{E}\}$ as a weighted bipartite graph, where $\mathbf{V} = \mathbf{X} \cup \mathbf{Y}$ is the vertex set, $\mathbf{E} = \{\omega_{ij}\}$ is the edge set, and ω_{ij} represents the probability value that y_j is just the next block of x_i in the reading order. The probability value ω_{ij} is computed by a comprehensive consideration as Formula 1.

$$w_{ij} = a \times PosN_{ij} + b \times CutN_{ij} + c \times RenderN_{ij} + d \times ContentN_{ij} + e \times StyleN_{ij} \quad (1)$$

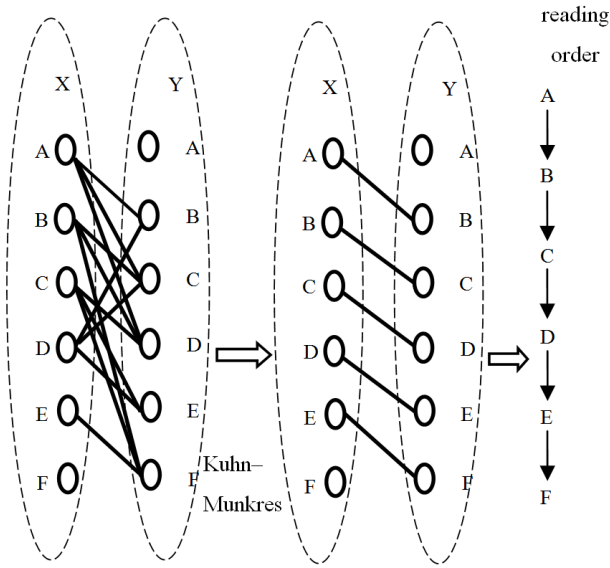


Figure 3: The bipartite graph of Figure 2 used in OM to get the reading order

Where $PosN_{ij}$ is the probability that block j is the next block of block i in position, and is calculated by the spatial reasoning method proposed in [1]. $CutN_{ij}$ is the probability that block j is the next block of block i in the “Cut-Order” of Step d) of Section 6.1.2.1. $RenderN_{ij}$ is the probability value that block j is output later than block i in the natural order of a PDF file. $ContentN_{ij}$ reflects the content continuity and similarity between block j and block i . For example, when the first word of block j and the last word of block i have the matched word order or part of speech, $ContentN_{ij}$ is assigned a high value. $StyleN_{ij}$ reflects the style similarity between block j and block i . For example, when block j and block i adopt the similar font, block width, alignment, $StyleN_{ij}$ is assigned a high value. a, b, c, d and e are respectively the weights of their connected probability, and $a + b + c + d + e = 1$. Their values are optimized on a training set and they are set to 0.15, 0.25, 0.20, 0.25, and 0.15 respectively.

An edge is constructed between block i and block j , when they are neighboring in position and w_{ij} is above a threshold. After the bipartite graph, $G = \{X, Y, E\}$ is constructed, each matching of the graph represents a possible reading order. And we employ the Kuhn-Munkres algorithm to compute the optimal matching of the constructed bipartite graph. Then the reading order is extracted from the optimal matching by finding the connecting y_j of each x_i . The bipartite graph of Figure 2 is shown in Figure 3.

6.1.3 Associating Figures and Captions

Associating a figure with its caption is important for the structured document generation and within-document navigation [7]. When a page contains more than one figure, such association may become challenging because of various layout relationships between figures and their captions. Lin et al. [21] propose a method to connect figures and their captions based on their distance and position in the vertical direction. However, the method assumes that the position of the caption candidates is all above or under the figures, and cannot process pages with varied relative positions between



Figure 4: An example with multiple figures and captions (P1, P2, P3 and P4 respectively represent the left, upper right and lower right figures. T1, T2, T3 and T4 are their corresponding captions.)

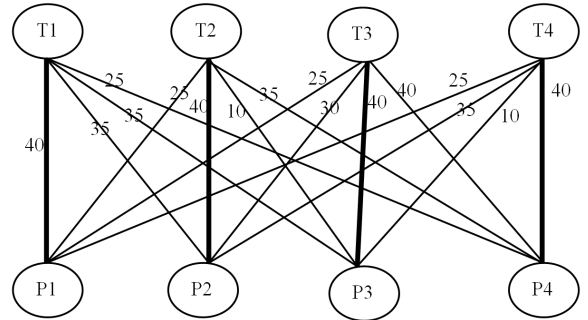


Figure 5: The bipartite graph of Figure 4 (the thick lines show the optimal matching and the determined relation between figures and captions.)

figures and their captions. For example, Figure 4 shows an example of multiple figures and captions on the same page. And Lin’s method cannot process this page.

Associating figures with their captions can be considered as a one-to-one matching problem. Thus we can also construct a weighted bipartite graph to model those figures, captions, and their association degrees, and then solve the OM problem through the Kuhn-Munkres algorithm.

The OM method for the task is similar to that described in Section 6.1.2 for the ordering problem. In the constructed bipartite graph, two parts of vertices respectively consist of figures and captions. And the weight of an edge from a figure to a caption is calculated according to their distance and overlapping degree. After executing the Kuhn-Munkres algorithm on the constructed bipartite graph, the relationship between figures and captions is extracted from the optimal matching of the graph. The

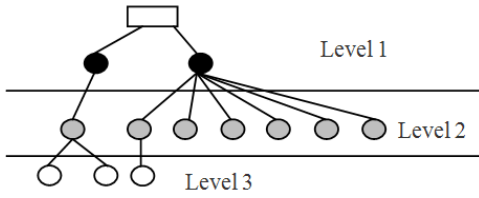


Figure 6: Tree structure of a ToC: the reading order is decided through depth-first traversal

bipartite graph of the example shown in Figure 4 and the optimal matching is shown in Figure 5.

6.2 Document Level Extraction

6.2.1 Hierarchy Structure Extraction

One goal of logical structure extraction at the document level is to determine the logical level of each chapter/section in a book. This goal is hardly achievable from ordinary book pages. But the ToC in books consists of index entries, each of which contains some functional items pertaining to a chapter or section, such as chapter title, chapter number, and page number reference. So analyzing the ToC pages can effectively extract the hierarchical structure.

In recent years, some rule-based methods of ToC entry level extraction have been proposed. For example, He et al. [17] combine geometrical rules (indentations) and semantic rules (typical text sequences identifying chapters and sections) to extract the hierarchical structure in Chinese books. Déjean et al. [11] decide the hierarchical organization of the ToC based on the assumption that ToC entries belonging to the same hierarchical level share the same visual characteristics and the entries at the lower level have more adjacent entries from the same level. In other words, ToC entries conform to SCC. But the entries with few siblings can potentially lead to incorrect level decision in their method. For example, their method cannot process cases shown in Figure 6 because the second level contains more adjacent entries (5) than the third level (2).

SEB uses the general containing relationships of ToC entries' levels to determine the ToC hierarchical structure: the lower level is contained in its adjacent and higher level. For example, the next entry of a chapter entry may be a chapter entry or a section entry, but not a subsection. The details of ToC hierarchical structure extraction are introduced in our previous paper [14]. After the hierarchy of ToC entries is determined, the chapters/sections are assigned the same hierarchies with their corresponding entries. When a book has no ToC pages, our system generates a ToC of the book by collecting the extracted headings in Section 6.1.1. Then the generated ToC is processed by the proposed hierarchy structure extraction method. Such technique can effectively expand the application space of the proposed method by covering books without ToC pages.

6.2.2 Metadata Extraction

The metadata of a book usually appear in the book's front and back parts. For example, the title of a book usually appears on the cover page, the publisher appears on the copyright page, and the citation data appear on the reference pages. This paper focuses on metadata extraction from the title page for their important role in automatic

metadata harvest and searching. The metadata extraction from citation data is discussed in our previous paper [12].

A number of methods on this task have been reported in recent years. And those approaches can be classified into three types: rule-based, learning-based, and template-based [10]. But they concentrate on a specific document type, namely scientific papers, and the metadata extraction from the title page of a book has not been reported in the existing literature according to our knowledge. This paper addresses this problem with a rule-based method.

The title page of a book usually includes four parts: abstract, catalog in publication (CIP), publisher-specific data, and copyright statement. We have categorized the metadata types of a book based on the analysis of many books. There are about 30 metadata types for Chinese books. Each type of the metadata includes some common keywords and differs from other types in position, style or character pattern. For example, in Chinese books the author field usually contains fewer than five words and its first word is the family name. The characters of ISBN are usually in a specific format. Based on the formal and content features of each metadata type, SEB creates a set of rules for each metadata type. And the system assigns each text block a probability of a metadata type after checking against the predefined rule set. In the initial result, each block might have high probability in multiple metadata types. In order to select only one metadata type for a block, the system also uses OM of bipartite graph to achieve a global optimal metadata assignment, which can be regarded as a one-to-one matching problem.

The OM for this task is similar to the method described in Section 6.1.2 for the ordering problem. In the constructed bipartite graph, two parts of vertices respectively consist of the text blocks in the title page and all predefined metadata types. And the weight value of an edge from a block to a metadata type is calculated from the rule-based processing. After executing the Kuhn-Munkres algorithm on the constructed bipartite graph, the assignment between the blocks and the metadata types is extracted from the OM.

7. EXPERIMENTS

To verify the performance of our method, we have done extensive experiments on various document collections. In this section we summarize the major findings.

The test data set consists of 1000 books in PDF format. The books are selected from a Chinese digital library⁴ and an electronic book store⁵. In order to measure the methods' applicability to larger collections, we select documents with various styles from 136 publishers. We list both recall rate (R) and precision rate (P) in the following experiments. We have also implemented the methods proposed by others (except for WISDOM++ [2]⁶, which is already a ready-to-use package) to compare with our methods. Given the features utilized, our methods are language-independent except for page element labeling, reading order detection and metadata extraction.

7.1 Physical Structure Extraction

⁴<http://www.apabi.com.cn/>

⁵<http://www.cnshare.org/>

⁶<http://www.di.uniba.it/~malerba/wisdom++/>

Table 2: Experimental results of global typography detection

Items	Precision (%)	Recall (%)
Body text font	100	100
Header and footer	99.09	99.07
Page body area	99.15	99.15
Text(String) line direction	100	98.62
Column	98.50	95.18
Line spacing of body text	100	100
Fonts used in headings	91.74	87.94

Table 3: Experimental results of detecting headers/footers

Methods	Precision (%)	Recall (%)
[22]	98.83	98.27
Our method	99.09	99.07
[19]	95.93	83.00

First, we measure the precision rate and recall rate of the global typographies extraction, as shown in Table 2.

Second, we compare our header/footer detection method with the two methods proposed in the literature ([22, 19]), which are considered the standard header/footer detection methods. Both the method of [22] and our method assume the SCC characteristic in headers/footers, while the method of [19] relies on specific visual clues, namely the horizontal lines on the top of pages. The test result is shown in Table 3.

As we can see, the method of [19] has a relative low recall rate, because many books do not have the required horizontal lines in the header/footer areas. On the other hand, this method does enjoy a high precision rate on documents with such horizontal lines. Compared with the last row, the results shown in the first two rows demonstrate excellent performance of SCC for extracting headers/footers. Most errors in two SCC-based methods are caused by headers/footers being too close to the main text. And our method outperforms the method of [22] because of the optimized page extraction strategy.

7.2 Logical Structure Extraction

7.2.1 Page Element Labeling

In the test of page element labeling, we have compared our method with WISDOM++, which is a well-known language-independent document image processing system on text block segmentation and classification. For our method, we have also evaluated the effect of clustering blocks before labeling them in our method.

As shown in Table 4, the identification of page elements is improved by the SCC-based clustering before labeling blocks. And WISDOM++ identifies components on image files, while our method operates on PDF files. Thus our method can use not only the layout information adopted by WISDOM++ but also the text and font information. Furthermore, our method utilizes the relationship between the elements belonging to the same component. And the test result indeed demonstrates our method performs better than WISDOM++ in identifying text components.

7.2.2 Reading Order Detection

For the document ordering problem, we measure the

Table 4: Experimental results of page element labeling (%)

Items		Head ing	Figure caption	Table caption	Foot Note
[2]	P	91.19	88.28	87.34	93.13
	R	95.14	92.43	93.12	94.21
Ours (without clustering)	P	92.09	88.72	87.48	93.42
	R	95.16	93.19	93.33	94.37
Ours (with clustering)	P	95.31	92.16	90.12	95.32
	R	97.82	96.91	98.34	95.85

Table 5: Experimental results of reading order detection

Methods	Precision(%)	
	page level	book level
Baseline	55.62	34.49
Ishitani [18]	91.82	78.28
Aiello [1]	93.19	79.78
Ours	96.93	91.19
Ours(without OM)	91.74	78.04
Ours(without natural order)	93.33	80.13

precision rates of different methods at both the page and book levels. A reading order of a page is considered correct when all the elements of the page are assigned in the right order. At the book level, we consider the reading order of a book correct when at least 95% of the pages in the book are correctly ordered. In this test, we select the natural rendering order of PDF files as the baseline. And we compare our method with the methods proposed by Aiello [1] and Ishitani [18]. In order to analyze the effects of different steps of our methods, we test our ordering method without using natural order in Step c) and Step e) of Section 6.1.2.1 or bipartite graph model in Step e) of Section 6.1.2. The test results are shown in Table 5.

The baseline result shows that the natural order sometimes agrees with the real reading order but it is not always reliable. So the natural order can serve as a valuable reference in the ordering process. As can be seen from the table, our OM-based ordering method achieves the highest precision rate. When our method does not execute the OM step, it actually degrade to a conventional XY-cut method and it achieves a precision rate similar to the method proposed by Ishitani. The results shown in Rows 5 and 7 prove the benefit of utilizing the PDF’s natural rendering order. Furthermore, compared with the method of [1], our method utilizes this natural order and extra content information, and a better result is achieved, as shown in Row 4 and 5. Most errors of our method occur in some vertically typeset PDF files generated from OCR software, in which the natural order is mostly wrong.

7.2.3 Associating Figures and Captions

For the task of associating figures with captions, we compare the method proposed by Lin [21] and our method and present the results in Table 6. The OM-based method not only uses the information of the figure and the associated caption, but also leverages the global information among all the figures and captions. Furthermore, our method is more versatile than Lin’s method, which assumes the caption is above or under the figure.

Table 6: Results of figure and caption association

Methods	Precision (%)	Recall (%)
Lin [21]	92.25	90.88
Our method	98.22	96.74

Table 7: ToC parsing experimental results

Tasks	Precision (%)	Recall (%)
Our method	94.91	94.91
[11]	91.13	91.13
[17]	62.28	62.28

7.2.4 Hierarchy Structure Extraction

The test results on ToC hierarchy recognition are shown in Table 7. We compare our ToC hierarchy determination method with those proposed in [11] and [17].

The method of [17] is only effective for books whose ToC entries have obvious indentations or standard chapter/section numbers. Unfortunately, many books do not satisfy such requirements and this explains the low recall rate of about 60%. Our method and [11] employs the SCC feature of ToC entries to identify ToC hierarchy and thus performs much better. Some books share the same format across two levels. In those cases our method and [11] incorrectly group the entries in the two levels into one cluster. Meanwhile, our method outperforms [11] by avoiding the errors caused by the entries in lower level with fewer siblings.

7.2.5 Metadata Extraction

We test our method on extracting metadata from book title page with and without OM step. 13 fields of metadata are taken into account. We use the following equation to measure the extraction accuracy. The experimental results are shown in Table 8. α is the extraction result of the rule-based metadata extraction method. β is the extraction result of the rule-based method plus the OM step.

$$Accuracy = \frac{\text{Number of correctly extracted fields}}{\text{Total number of fields}} \times 100\% \quad (2)$$

From Table 8, it can be seen that the OM step benefits metadata extraction in every metadata type. Several metadata types, such as publisher, date, ISBN, URL, have distinguishing characteristics, and thus a high accuracy is achieved. Other metadata fields, such as price, paper size, word number, have similar features. And the OM step improves the results even more in such fields.

The program is implemented in C++ and the experiments are conducted on a standard PC (dual-core 1.4GHz, 2GB RAM, Windows Server 2008). The total executing time of all tasks on a book with 300 pages is about 34 seconds.

8. CONCLUSIONS

In this paper, we introduce a system, SEB, to extract structure information from book documents in PDF format. The system extracts the major physical and logical structures of a book at the page and document levels, which include global typographies, logical text components, reading order, hierarchical structure of chapters/sections, and metadata. In the context of PDF-based books, we propose a set of new methods to improve the conventional image-based methods by using the two characteristics of PDF files and books: style consistency of page component

Table 8: Results of metadata extraction from title page

Items	α (%)	β (%)
Author	92.56	95.14
Editor	90.59	94.42
Title	95.17	95.56
Publisher	98.31	98.63
Date	96.53	96.97
Price	91.84	93.45
Issue	87.27	90.22
ISBN	98.52	98.83
Pager size	88.33	91.52
Word number	91.45	93.48
Telephone	93.28	93.68
Address	92.66	93.86
URL	97.43	97.63

(SCC) and local reliability of natural rendering order. And optimal matching (OM) of bipartite graph is used for the first time to determine the reading order, to match figures and their captions, and to extract metadata from title pages. Experimental results show that the proposed methods perform better than the traditional image-based methods. The system has significantly improved the productivity of an e-book producing pipeline by automating the structure extraction step. In addition, the proposed method has been incorporated into a commercial electronic book editing software package⁷ and about 30,000 books have been processed by the system in the past year. The finished books with logical structure can be read on iPad using Apabi reader⁸, which can be downloaded from Apple's App Store. The extracted structure information enables the reflowed reading mode in this app.

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⁷<http://www.apabi.cn/product.shtml>

⁸<http://www.apabi.cn/html/201011/1011191606123494.shtml>

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